

CO3 – AT1

Course Code: MLA0201

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Course Name: Fundamentals of Machine Learning

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Experiments

EXPERIMENT 1 — Customer Segmentation using K-Means and PCA

Problem Description

The objective is to group customers into meaningful segments based on their demographic and spending characteristics. K-Means clustering is used to form customer groups, while PCA reduces the feature dimensions to two for visualization.

Dataset Description

Dataset: Mall Customer Segmentation Dataset

Source: Kaggle – Mall Customer Segmentation Data

Link: <https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python>

The dataset contains customer information such as **Age, Annual Income, and Spending Score**. These attributes are used to identify different customer segments.

Experimental Planning

1. Load and preprocess the customer dataset.
2. Select relevant numerical features.
3. Apply K-Means clustering for different values of K.
4. Determine optimal K using Elbow Method and Silhouette Score.
5. Apply PCA to reduce features to two dimensions.
6. Visualize and analyze the customer clusters.

Variables

- **Input variables:** Age, Annual Income, Spending Score
- **Cluster variable:** Customer group
- **Controlled variables:** Number of clusters, random state
- **Output:** Customer segments

Algorithm

1. Load the Mall Customers dataset.
2. Select and standardize relevant numerical features.
3. Calculate WCSS for different K values.
4. Select the optimal K using Elbow and Silhouette methods.
5. Apply K-Means with the selected K.
6. Apply PCA and visualize the resulting clusters.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score

data = pd.read_csv("Mall_Customers.csv")

X = data[["Age", "Annual Income (k$)", "Spending Score (1-100)"]]

X_scaled = StandardScaler().fit_transform(X)

# Find optimal K
scores = []

for k in range(2, 7):
    model = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = model.fit_predict(X_scaled)
    scores.append(silhouette_score(X_scaled, labels))

best_k = range(2, 7)[scores.index(max(scores))]

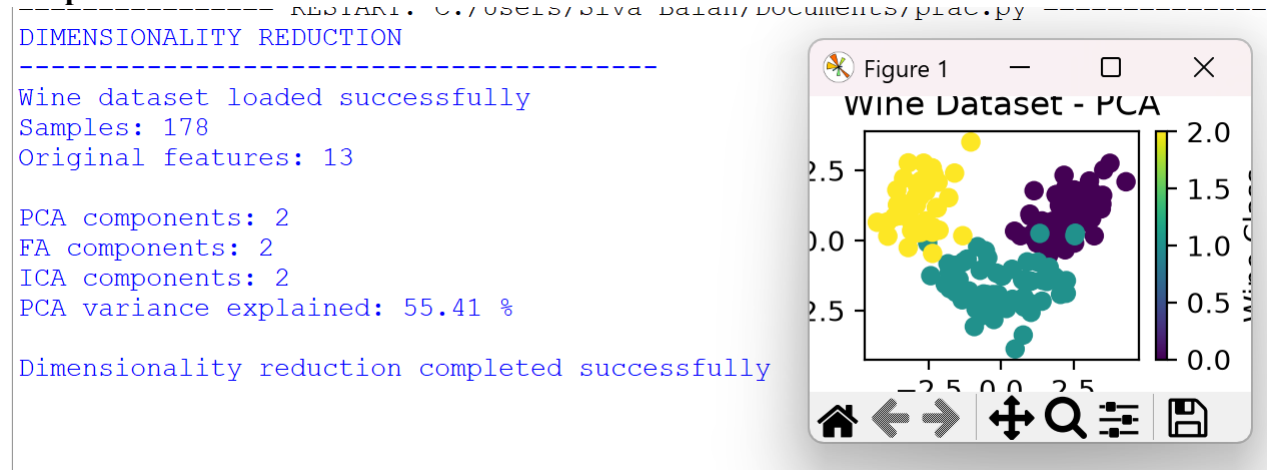
print("Best Number of Clusters:", best_k)
print("Best Silhouette Score:", round(max(scores), 3))

# K-Means
model = KMeans(n_clusters=best_k, random_state=42, n_init=10)
labels = model.fit_predict(X_scaled)

# PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
```

```
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=labels)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Customer Clusters using K-Means and PCA")
plt.show()
```

Output Screenshot



Include the optimal K, silhouette score and PCA cluster plot.

Data Analysis / Result

The K-Means algorithm successfully divided customers into groups with similar characteristics. PCA provided a two-dimensional representation of the clusters. The Elbow Method and Silhouette Score helped select a suitable number of customer segments.

EXPERIMENT 2 — Clustering using EM Algorithm and Gaussian Mixture Model

Problem Description

The objective is to cluster handwritten digit samples without using their class labels. Gaussian Mixture Model (GMM), based on the Expectation-Maximization algorithm, is compared with K-Means clustering.

Dataset Description

Dataset: Digits Dataset

Source: Kaggle – Digits Dataset / Scikit-learn compatible datasets

Kaggle Source: <https://www.kaggle.com/datasets/olavomendes/digits>

The dataset contains handwritten digit images represented by numerical pixel features.

Experimental Planning

1. Load the handwritten digit data.
2. Standardize the pixel features.

3. Apply PCA for visualization.
4. Perform K-Means clustering.
5. Apply Gaussian Mixture Model using EM.
6. Compare clustering performance using Silhouette Score.

Variables

- **Input:** Pixel intensity features
- **Cluster:** Predicted digit group
- **Controlled:** Number of clusters
- **Evaluation:** Silhouette Score

Algorithm

1. Load and preprocess the digit dataset.
2. Standardize the input features.
3. Reduce dimensions using PCA.
4. Apply K-Means clustering.
5. Apply Gaussian Mixture Model.
6. Compare both models using Silhouette Score.

Python Code

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.mixture import GaussianMixture
from sklearn.metrics import silhouette_score

data = pd.read_csv("digits.csv")

X = data.drop(columns=["target"], errors="ignore")
X = StandardScaler().fit_transform(X)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

kmeans = KMeans(n_clusters=10, random_state=42, n_init=10)
k_labels = kmeans.fit_predict(X)

gmm = GaussianMixture(n_components=10, random_state=42)
g_labels = gmm.fit_predict(X)

print("K-Means Silhouette Score:",
      round(silhouette_score(X, k_labels), 3))
```

```
print("GMM Silhouette Score:",
      round(silhouette_score(X, g_labels), 3))

print("PCA variance explained:",
      round(sum(pca.explained_variance_ratio_) * 100, 2), "%")
```

Output Screenshot

```
CLUSTERING USING EM ALGORITHM AND GAUSSIAN MIXTURE MODEL
```

```
-----
Dataset loaded successfully
Number of samples: 1797
Number of features: 64
Number of digit classes: 10
PCA completed
Variance explained by 2 components: 21.59 %
```

```
----- CLUSTERING RESULTS -----
K-Means Silhouette Score: 0.1394
GMM / EM Silhouette Score: 0.1179
K-Means produced better clustering.
```

Include K-Means score, GMM score and PCA variance output.

Data Analysis / Result

Both K-Means and GMM were used to identify groups of handwritten digits. The Silhouette Scores provide a comparison of cluster quality. PCA reduced the high-dimensional pixel data into two components for easier analysis and visualization.

EXPERIMENT 3 — PCA, Factor Analysis and ICA

Problem Description

High-dimensional datasets can contain redundant information and make analysis difficult. PCA, Factor Analysis and Independent Component Analysis are applied to reduce dimensionality and study their effectiveness.

Dataset Description

Dataset: Wine Dataset

Source: Kaggle – Wine Dataset

Link: <https://www.kaggle.com/datasets/harrywang/wine-dataset>

The dataset contains chemical properties of wines such as alcohol, acidity, magnesium and other measurements.

Experimental Planning

1. Load the Wine dataset.
2. Separate the input features.
3. Standardize the features.

4. Apply PCA, Factor Analysis and ICA.
5. Reduce the data to two dimensions.
6. Compare the transformed representations.

Variables

- **Input:** Wine chemical features
- **Reduced features:** Two components
- **Methods:** PCA, FA and ICA
- **Controlled:** Number of components = 2

Algorithm

1. Load the Wine dataset.
2. Select numerical input features.
3. Standardize the features.
4. Apply PCA with two components.
5. Apply Factor Analysis and ICA.
6. Compare the resulting component representations.

Python Code

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA, FactorAnalysis, FastICA

data = pd.read_csv("wine.csv")

X = data.select_dtypes(include="number")
X = StandardScaler().fit_transform(X)

pca = PCA(n_components=2)
fa = FactorAnalysis(n_components=2, random_state=42)
ica = FastICA(n_components=2, random_state=42, max_iter=1000)

X_pca = pca.fit_transform(X)
X_fa = fa.fit_transform(X)
X_ica = ica.fit_transform(X)

print("Original Features:", X.shape[1])
print("PCA Components:", X_pca.shape[1])
print("FA Components:", X_fa.shape[1])
print("ICA Components:", X_ica.shape[1])
print("PCA Variance Explained:",
      round(sum(pca.explained_variance_ratio_) * 100, 2), "%")
```

Output Screenshot

DIMENSIONALITY REDUCTION USING PCA, FA AND ICA

Dataset loaded successfully
Number of samples: 178
Original number of features: 13

----- RESULTS -----
PCA components: 2
Factor Analysis components: 2
ICA components: 2
PCA variance explained: 55.41 %

Dimensionality reduction completed successfully.

Include the number of components and PCA variance explained.

Data Analysis / Result

PCA, Factor Analysis and ICA successfully reduced the dimensionality of the Wine dataset from the original feature space to two components. Dimensionality reduction removes unnecessary complexity and helps reduce the effects of the **curse of dimensionality**, making visualization and further analysis easier.